INSTITUTE OF HEALTH INFORMATICS

Graduate Programme in Health Data Science

Assessed Coursework Submission

Student candidate number:	WFDH2
Module:	CHME0021: Dissertation proposal in Health Data Science
Date due:	29 April, 16:00 BST
Word count: (excluding references, diagrams and appendices)	2000
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INSTITUTE OF HEALTH INFORMATICS

Dissertation Proposal:
MSc Health Data Science 2025-2026

Candidate No: WFDH2

Instructions: *Please note that while all sections should be included, the word counts for each section are for guidance only and the exact distribution may vary from project to project depending on its focus. Its optional if you want to use this template for your submission, but all submissions should include the below headers*

You may only use generative AI as a spell or grammar checker for your write-up. Some projects may use generative AI in an assistive role for their analysis, which can be allowed if in agreement with the supervisor. Any use of AI has to be fully described in a statement in the proposal and thesis. For the proposal please write this statement at the end of the document before the reference list.

1) Project Title & 3-6 Keywords		
"Text Mining PubMed: Mapping Treatment Evolution and Adverse Events in Cystic Fibrosis" Keywords: Text Mining; PubMed; Cystic Fibrosis; Adverse Events; Network Analysis; Temporal Analysis		
2) Project aims, objectives and rationale		Words
	Please define concisely the aims / research questions / hypotheses of the proposed project in data science for health & biomedicine.	200

Despite PubMed's comprehensive coverage of the scientific literature, it is an underutilized source for analysing changes over time in medical research. Using text mining and network analysis, this project will examine longitudinal changes in scientific language on the treatment of cystic fibrosis (CF) and associated adverse events, focusing on vocabulary. Text mining and network analysis will be applied to bibliographic data to examine how treatments and adverse events associated with cystic fibrosis have changed from 1970-2024, with an emphasis on treatment and adverse event-related terminology. Its key objectives are:

- Extracting cystic fibrosis within titles/abstracts from PubMed using search strings for pre-processing
- Identifying treatment and adverse event-related terms within titles/abstracts using pre-defined clinical search terms from approved CFTR modulators (CF-specific treatments)
- Quantifying vocabulary richness using *Gini coefficient*, *Yule's K*, and *Hapax Legomenon*
- Mapping associations between treatments and adverse event outcomes using network analysis
- Analysing quinquennial changes in treatment and adverse event co-occurrence networks and vocabulary richness from 01/01/1970 to 31/12/2024

3) Introduction

Words

Please describe the background, context, and importance of the proposed research in light of related literature. Show a sound understanding of the problems presented in the data.

550

Cystic Fibrosis (CF) is a genetic disease caused by a faulty version of the CFTR gene. The movement of salt and water in and out of cells is controlled by the CFTR gene and the protein it makes. If both the gene and protein are faulty, it causes thicker mucus production, which affects multiple organs. Primarily, the lungs are negatively affected by thick mucus blocking airways, causing respiratory infection, organ damage, and shortening life expectancy (UK Cystic Fibrosis Registry, 2024).

The clinical course of CF has drastically evolved over the decades. Historically, CF was known as a fatal childhood disease, with patients in the 1960s rarely surviving past 5 years of age (Scotet et al., 2020). However, current treatments, ranging from advanced symptomatic management to CFTR modulators, have significantly increased survival, with today's estimated median survival age being 66 years in high-income countries (UK Cystic Fibrosis Registry, 2024). Therefore, a better understanding of CF's long-term disease management, treatments, and adverse outcomes is essential to improving survival.

PubMed is a free, publicly available database containing over 39 million citations and abstracts of scientific and medical research. It provides titles, abstracts, medical subjects headings, and publication dates, allowing for a systematic way to text mine and analyse trends in the scientific literature of CF. It is important to note that English is the primary language of scientific journal publications (approx. 95.7%), which allows for simpler, consistent language for text mining (Karcioglu et al., 2025).

After the discovery of the CFTR gene in 1989, CF treatment has shifted from a disease of symptom management (e.g., antibiotics and nutrition support) to a targeted molecular therapy, introducing CFTR modulators and combination therapies such as elexacaftor/tezacaftor/ivacaftor (Lopes-Pacheco, 2020; Scotet et al., 2020).

CF-related literature is growing at a considerable rate. Consequently, the need for efficient methods of identifying patterns across multitudes of studies grows. By analysing literature, treatment and adverse event associations can be mapped to see how they have evolved, giving insight into long-term trends that are not observable through traditional reviewing methods. Clearer, more visible trends can be identified, thus contributing to a firmer understanding of new, emerging therapies alongside the overall long-term safety of these therapies.

Biomedical databases, such as PubMed, greatly accelerate the volume of newly available literature in the biomedical domain, with MEDLINE (PubMed's database) exceeding 23 million indexed articles in 2017 (Renganathan, 2017). The vast amount of literature makes it time-consuming to extract knowledge manually. Therefore, there is a need for scalable, reproducible automation of extracting information from text. Text mining–a process of extracting meaningful information from unstructured text–uses automated methods like tokenization and frequency analysis to create a systematic way to identify changes in the language of treatment and adverse events more efficiently (Silge & Robinson, 2016). It provides a large-scale means of quantifying vocabulary diversity and identifying patterns in treatment and adverse event language from biomedical literature (Renganathan, 2017; Harpaz et al., 2014). Network analysis is subsequently utilized to represent the quantified terms as nodes, and the co-occurrence frequency as edges, illustrating the associated relationships between treatments and adverse events over time (Al-Aamri et al. 2019).

4) Methodology

Words

Please describe the proposed methods of analysis (statistical or other appropriate methods of health data science). Include a brief description of the dataset which will be evaluated to meet the study objectives.

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Data Source and Search Strategy:

PubMed as the Primary Data Source
This project uses PubMed as its primary data source. PubMed contains a comprehensive repository of biomedical literature with titles/abstracts [tiab] and publication dates.

Search String Justification
Manually created search strings used free-text terms. Two filters were applied:

- **Title/Abstract [tiab]** - captures language and terms explicitly in the text
- **English** - only the English language was applied to the searches for consistent language for downstream n-gram frequencies (most CF literature is in English)

The following search strings were used as the basic search:

- **"cystic fibrosis"[tiab]** - specific disease of interest
- **CFTR modul*[tiab]** - screens publications containing CFTR modulators as they are CF-specific treatments; right truncation applied to include variations of the word

Boolean operators (OR, AND) and truncation (*) were used to refine and expand searches.

Date Range
The publication date range was constrained to 01/01/1970 to 31/12/2024 using the [dp] field tag.

- **"1970/01/01"[dp] : "2024/12/31"[dp]**

Although CF was first described by Dorothy H. Anderson in 1938 (Scotet et al., 2020) and the first recorded abstract containing the term "cystic fibrosis" was indexed in PubMed in 1956, abstract counts prior to 1970 were scarce. Thus, 01/01/1970 was selected as the start date. The end date of 31/12/2024 was chosen to include the most recent literature, broken down into 5-year periods to avoid partial years (i.e. exclude 2025 onward), ensuring 11 consistent 5-year intervals from 1970-2024 for the analysis of temporal trends in CF literature.

Excluding MeSH Terms
Medical subject headings [MeSH] were not included in the search string criteria because of its lack of historical coverage of abstracts pre-1970s and its nature to identify papers by concept rather than wording, which is essential for text mining and network analysis.

Corpus Construction
All text mining analysis is done using the R programming language. Using the `pubmedR` package (Cuccurullo, 2017), the parent corpus consisted of 46,244 results as of 20/04/2026, with subsets for organs, treatments, adverse events, and life expectancy . Each result contained a title, abstract, and publication date. For pre-processing, steps will include:

- **Tokenization** - cuts up text into smaller units (tokens) so a computer can process unstructured text
- **Lowercasing** - converts all characters in the text to lowercase; reduces dimensionality of text data
- **Normalization**
  - **Stemming** - reduces words to their common stem (e.g., studies → studi); very fast, less accurate
  - **Lemmatization** - reduces words to their base form (e.g., studies → study); accurate, but time-consuming
- **Stopword Removal** - remove words that don't contribute meaningful interpretation; often are high frequency and disrupt computational processes
  - Removal of stop words will use the package `tidytext`'s `stop_words` dataset (Silge & Robinson, 2016)

Analytical Methods:

Term Identification and Frequency Analysis
Treatment and adverse event-related terms will be identified using n-gram frequency analysis, where word sequences from the text will be extracted and filtered only if the word sequences are recurring above a certain threshold (Yip et al., 2010). Thus, only meaningful terms will be kept. Log-odds ratios will then be used to identify which terms are disproportionately associated with specific quinquennial periods (Pojanapunya et al., 2018).

Vocabulary Richness Analysis
Linguistic diversity will be quantified using:

- *Gini Coefficient* - measures inequality in word frequency distribution

- Originally used to measure income inequality distributions (Gini, 1912, cited in Ceriani, L. and Verme, P. 2011), but has now been applied to vocabulary distribution analysis as well (Popescu, 2009)
- *Yules K* - assesses lexical diversity and repetition (Tanaka-Ishii et al., 2015)
- *Hapax Legomena* - captures words that occurred only once (Tweedie, F.J. & Baayen, R.H., 1998)

Network Analysis
Co-occurrence networks will be constructed. Nodes will represent terms, and edges will represent their co-occurrence within publications. This allows the relationship between treatments and adverse events to be identified, as well as key clusters within the literature (Newman, 2010).

Temporal Analysis
The dataset will be stratified quinquennially from 1970-2024, applying each analytical tool–term frequency, vocabulary richness, and network analysis– to examine longitudinal trends separately in each period.

5) Results

Words

Expected results, define clearly what is expected to be achieved by the end of the proposed project. You can add pilot results, if available at this stage.

300

Preliminary findings from PubMed queries and expected results:

Corpus Size & Feasibility

1. From 1970-2024, PubMed returned 43,760 results for ("cystic fibrosis"[tiab] OR CFTR modul*[tiab]) AND ("1970/01/01"[dp] : "2024/12/31"[dp]). Therefore, text mining is feasible with this corpus size. Early records before the 1970s lacked a sufficient number of abstracts. Thus, only records from 1970-2024 were retrieved, divided into 11 quinquennial periods.

Temporal Trends

2. Exploratory results indicate an increase in CF literature over all quinquennial periods (Figure 1). Lung-related research represents the largest category, initially containing 21 publications from 1970-1974 to 5,886 publications from 2020-2024. Liver and pancreas-related literature don't appear as frequently but show steady growth over time. Treatment-related publications show a significant rise from 2010 onwards.

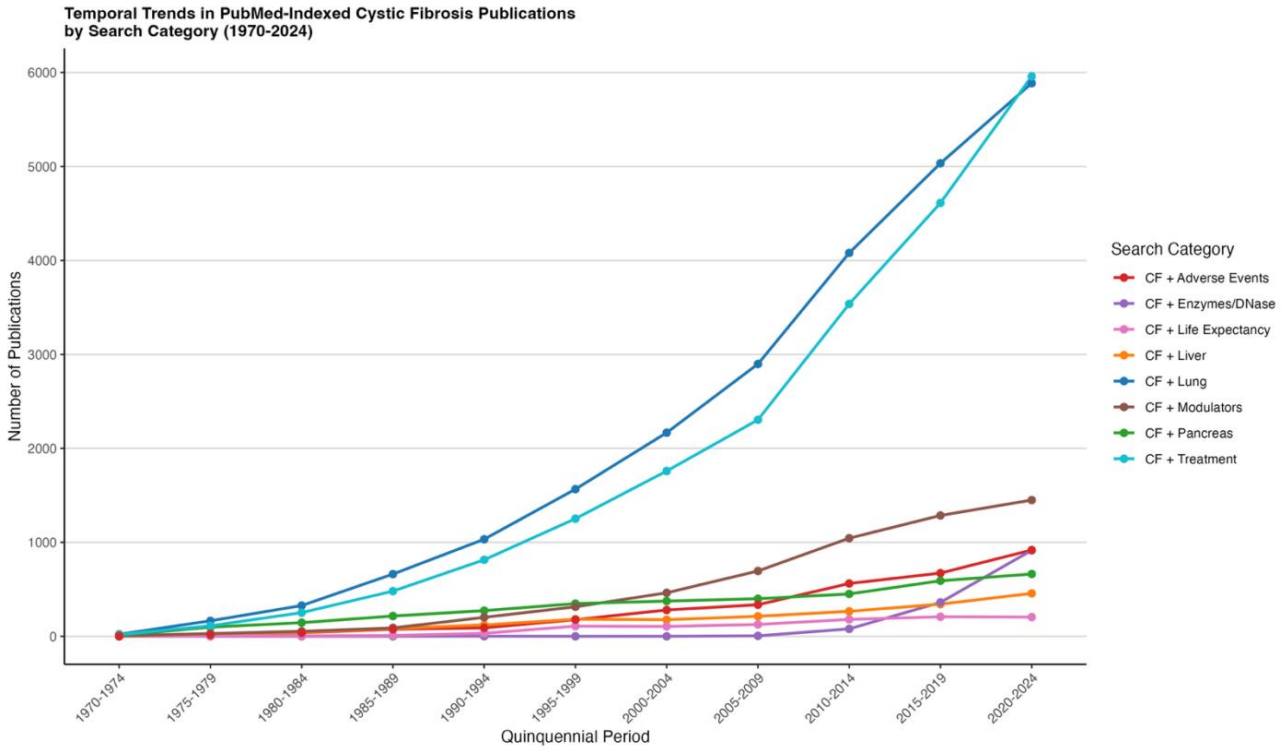


Figure 1. Temporal trends in the number of PubMed-indexed cystic fibrosis publications by search category across 11 quinquennial periods (1970-2024). Each line pertains to a specific category combining CF or CFTR modulator terms with category-specific keywords. The number of publications was retrieved using a PubMed search string, filtered by title and abstract fields [tiab].

Treatment and Adverse Event Trends

3. CFTR modulators appear in more recent decades, showing a clear increase from 2010-2014 onwards, rising from 79 to 915 publications by 2020-2024. This can be attributed to newer literature focusing on precision medicine therapies (e.g., ivacaftor) instead of symptomatic treatments. Adverse event-related publications show steady increases from 1970-2024, with an increase from 2010 onwards, likely reflecting drug safety requirements because of the introduction of CFTR modulators.

Expected Outputs

4. Future analysis will generate term frequency distributions, temporal trend visualizations, and network graphs with associations mapped between treatments and adverse events.

6) Discussion	Words
Expected discussion points. Describe what the proposed research adds to current scientific evidence, and how it contributes to planning and/or providing better health services. Include a fair assessment of the value of the planned analysis/results and a clear statement of the limitations.	300

This project aims to map associations between treatments and adverse event-related vocabulary using text mining and network analysis, quantifying and tracking changes in the linguistic diversity of CF research vocabulary over time. Through text mining and network analysis of PubMed data, this project creates a different approach to identifying trends in treatment and adverse events across decades of research.

The findings are expected to provide a clearer image of the shift from symptomatic management of CF to more precise, targeted molecular therapies like CFTR modulators. These therapies are associated with off-target effects. Areas of interest include mental health outcomes such as depression and anxiety (Talwalkar et al., 2017). With the current lack of understanding and characterisation of these effects, mapping associations between treatment and adverse events could potentially provide insight into unanticipated relationships between new therapies and their potential risks.

These trends have practical implications for clinical research priorities. An example of this can be seen in the increase of adults with CF with an estimated median survival age of 66 years (UK Cystic Fibrosis Registry, 2024). This data highlights a growing need for research on long-term management strategies. Gaps in the literature concerning how treatment and adverse event terminology have evolved together over time require a deeper longitudinal text-based exploration.

There are notable limitations. Firstly, the depth of identified relationships is limited because the analysis is on titles and abstracts, not the full text. Therefore, relevant information may be missing. Secondly, English language publications may introduce geographic bias. This could impact abstract availability, especially for earlier records that aren't in English. Publication bias exists particularly regarding advanced therapeutic developments being less prevalent and accessible to low and middle-income countries (Guo et al., 2024). Lastly, although we can map associations, this does not indicate causal relationships because text mining and network analysis are correlational. The co-occurrence of terms in text does not equate to directionality (e.g., "liver" does not distinguish between a positive or negative outcome); sentiment analysis can address this, however. Depending on the progress of other project elements, further investigation into sentiment analysis may be included.

Ultimately, longitudinal data-driven methods, including text mining, represent a novel approach for identifying adverse events associated with novel therapies at scale.

7) Ethical issues	Words
Please provide a summary of ethical issues and implications of the proposed project. Please consider this section even if the project is exempt from formal ethical approval.	100

The dataset used (PubMed) is a publicly available data source. No patient data, personally identifiable information, or confidential material is used. Additionally, the study did not require any ethical approval as there are no human participants in this project.

8) Statement on the use of Artificial-Intelligence based tools

Words

Please provide a statement that describes the use of AI-based tools during the dissertation project activities (including during and for the preparation of this proposal).

AI-based tools (Claude, ChatGPT, Grammarly) were used in an assistive role, limited to:

- Providing improving sentence clarity, grammar checks, and spelling checks to written work
- Conducting outside literature review/suggesting citations for methodological steps

All written work, coding, analysis, methodology and interpretation is original work. AI responses were critically reviewed to ensure accuracy, and no AI tools were used to generate data, conduct analysis, or produce results.

9) References

Words

Please provide a list of references

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